Kyle Christian Chong

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EDUCATION

University of Texas at Austin | GPA: 3.47 **Graduation: Spring 2027**
B.S. Mechanical Engineering

EXPERIENCE

Nanoscale Manufacturing and Design Laboratory - Undergraduate Researcher **January 2026 - Present**

- Designed LabVIEW-based automation for a precision optomechanical system, characterizing positioning repeatability on a 3-stage precision optomechanical system (CMOS, mask, aperture).
- Troubleshooted recurring calibration drift using root-cause analysis to ~5 °C lab-temperature swings (fishbone / Six Sigma), then instrumented DAQ-based temperature monitoring with a warning threshold to catch the condition before it skewed readings.
- Cleanroom-trained (contamination control, gowning); shadowed fab operations, including etch parameter tuning to correct groove misalignment

Imagine Devices - Associate R&D Engineer **March 2026 - Present**

- Designed and built a benchtop neonatal breathing simulator to validate a high-sensitivity pressure-sensing system, integrating sensor, DAQ, and fixturing into a repeatable test setup.
- Characterized the full pressure-sensor measurement chain end-to-end, quantifying combined uncertainty of ±0.05395V and establishing the error budget the team uses to trust device readings.
- Built the sensor calibration procedure and fixturing now used directly by another engineer for routine calibration.

ENGINEERING PROJECTS

Articulated Mechatronic Wings **May 2026 - Present**

- Synthesized 7-bar pantograph linkage that unfolds a 4.5-ft wing from a single actuator, using a custom Python solver to reduce 135 candidate solutions to the final mechanism.
- Sized actuator by deriving a quasi-static holding torque of 3.9 N·m per wing (7.8 N·m total) via the Principle of Virtual Work, using SolidWorks mass properties.
- Selected materials via a weighted decision matrix, specifying Carbon Fiber (CFRP) for high-shear central links and balsa (4× lighter) for outer links to maximize strength-to-weight.

Snap-Detecting Watch: Hardware & Enclosure Design **May 2025 - June 2026**

- Architected a 4-layer wearable PCB (KiCad) around an nRF52833, strategically isolating 2.4GHz RF routing from continuous-listening analog microphone traces to ensure high-fidelity acoustic signal integrity.
- Performed ANSYS structural analysis to quantify PCB stiffener effects, driving four physical design-test iterations that implemented analog filtering to eliminate enclosure-induced resonance and acoustic signal noise.
- Designed thin-walled enclosure revisions in SolidWorks, executing rigorous material-specific tolerance analysis (PLA/Resin) to ensure robust packaging for the battery, sensors, and PCB across multiple builds.

Snap-Detecting Watch: Real-Time Machine Learning Classifier **May 2025 - June 2026**

- Engineered an embedded DSP pipeline to extract time-domain acoustic features (ZCR, Crest) and PCA-reduced spectrograms from 2048-sample audio windows to prepare data for real-time classification.
- Evaluated classification models via exhaustive feature selection and hyperparameter grid searches, benchmarking linear against non-linear decision boundaries using a dataset of 300+ custom gesture samples.
- Trained a Random Forest model to 92.3% accuracy, prioritizing high trigger recall and minimal false-positives to ensure reliable gesture detection natively on an nRF52833 wearable device.

TECHNICAL SKILLS/CERTIFICATIONS

- Six Sigma/Lean Yellow Belt Coursera Certification
- CAD/FEA/Fab: SolidWorks, ANSYS, 3D printing, hand reflow/drag soldering
- Test / Prototyping: LabVIEW, NI DAQ/data acquisition, oscilloscopes, digital multimeters, thermocouples, calipers
- Data Analysis (Cleaning + Visualization): Excel, Python (NumPy, Pandas), MATLAB, PowerPoint, Word
- Embedded: KiCad, STM32CubeIDE (C) , analog sensors, digital signal processing